Pranav M

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Education

2023 – 2027 Electronics and Communication Engineering at **PES University**
2022 Class 12th at **Vydehi School of Excellence**

Projects

Reconfigurable DDR5 Memory Controller (RMC) GitHub
Architected a parameterized DDR5 memory controller targeting JEDEC JESD79-5 and DFI 5.2, with an AXI4 host interface and scalable multi-channel/multi-rank support. Defined microarchitecture across the client interface, transaction tracking and ordering, MC-core datapath, APB/CSR interface, and DDR5 mode-register and power-management flows; contributed read-path verification and a DFT observation interface.

HADES – Programmable PNM Engine GitHub — arXiv preprint
Designed and verified a programmable processing-near-memory engine for binary similarity search, featuring a 12-opcode ISA and dual-row/selective-scan mechanisms around a shared register-file macro. Submitted for tapeout through Tiny Tapeout; post-layout implementation closes timing at 40 MHz with zero reported DRC/LVS violations.

Aletheia – Processing-Near-Memory Framework GitHub
Built a PNM framework on a Raspberry Pi 5, dispatching commands over TCP to a memory node to execute computation locally. Ran PMU-backed characterization of working sets, stride patterns, and dataset scaling to study memory locality and cache behavior.

Herald – Fixed-Point DSP Coprocessor GitHub
Designed and taped out a Q12.12 DSP coprocessor through Tiny Tapeout, featuring CORDIC and multiply-accumulate (MAC) engines for FPU-less microcontrollers. Verified using Bluespec and cocotb.

Aetheron – RISC-V SoC GitHub
Built a simulation-based RISC-V SoC in Bluespec with a pipelined RV32I core, TileLink-lite interconnect, boot-from-ROM loading, and memory-mapped UART, GPIO, and timer peripherals; ran compiled C programs.

Experience

PES University – Teaching Assistant, Computer Organization & Design Aug 2026 – Present
Teach RISC-V assembly and processor microarchitecture, and guide students through the design, implementation, simulation, and debugging of single-cycle RISC-V processors.

HSP, PES University – Mentor Jul 2025 – Present
Mentored students on digital hardware design projects across two consecutive terms, guiding architecture design, reviewing HDL, and debugging design modules.

Skills

Languages	SystemVerilog, Verilog, Bluespec, Veryl, C, Python, Assembly
Tools & Platforms	Vivado, GTKWave, Verilator, Icarus Verilog, cocotb, ModelSim, Git, Linux
Domains	Computer Architecture, Memory Systems, DDR/Memory Interfaces, Processing-Near-Memory, RTL Design and Verification, SoC Design